

Erik Jonsson School of Engineering and Computer Science

Engineering (BA)

Bachelor of Arts in Engineering

[Degree Requirements](#) (123 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

The Bachelor of Arts in Engineering faculty includes faculty members from all departments within the Erik Jonsson School of Engineering and Computer Science.

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

[RHET 1302](#) College Writing II¹

[ECS 2390](#) Professional and Technical Communication¹

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2413](#) Differential Calculus²

or [MATH 2417](#) Calculus I²

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[PHYS 2325](#) Mechanics³

[PHYS 2326](#) Electromagnetism and Waves³

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[MATH 2414](#) Integral Calculus²

or [MATH 2419](#) Calculus II²

[PHYS 2125](#) Physics Laboratory I³

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 78 semester credit hours

Major Preparatory Courses: 14 semester credit hours beyond Core Curriculum

[ECS 2390](#) Professional and Technical Communication¹

[MATH 2413](#) Differential Calculus²

or [MATH 2417](#) Calculus I²

[MATH 2414](#) Integral Calculus²

or [MATH 2419](#) Calculus II²

[PHYS 2325](#) Mechanics³

[PHYS 2125](#) Physics Laboratory I³

[PHYS 2326](#) Electromagnetism and Waves³

[PHYS 2126](#) Physics Laboratory II

[RHET 1302](#) College Writing II¹

Choose 4 semester credit hours from the following:

[ECS 1100](#) Introduction to Engineering and Computer Science

[ECS 1210](#) Foundations in Engineering and Computer Science I

[BMEN 1100](#) Introduction to Bioengineering I

[CS 1200](#) Introduction to Computer Science and Software Engineering

[EE 1100](#) Introduction to Electrical and Computer Engineering

[MECH 1100](#) Introduction to Mechanical Engineering I

[ECS 1220](#) Foundations in Engineering and Computer Science II

[BMEN 1208](#) Introduction to Bioengineering II

[EE 1202](#) Introduction to Electrical and Computer Engineering II

[MECH 1208](#) Introduction to Mechanical Engineering

Additional Math/Science (9 semester credit hours minimum)

[ENGR 3341](#) Probability Theory and Statistics

Math (3 semester credit hours):

[ENGR 2300](#) Linear Algebra for Engineers

[MATH 2420](#) Differential Equations with Applications

[ENGR 3300](#) Advanced Engineering Mathematics

Science (3 semester credit hours):

[CHEM 1311](#) General Chemistry I

[CHEM 1111](#) General Chemistry Laboratory I

[BIOL 2311](#) Introduction to Modern Biology I

[BIOL 2111](#) Introduction to Modern Biology Workshop I

Major Core Courses: 27 semester credit hours

Design/Experimental Methods (2 semester credit hours)

[BMEN 3200](#) Biomedical Engineering Fundamentals and Design

[CS 3354](#) Software Engineering

[EE 2310](#) Introduction to Digital Systems

[MECH 3305](#) Computer Aided Design

[MECH 3105](#) Computer Aided Design Laboratory

CS Programming (3 semester credit hours)

[BMEN 1300](#) Introduction to Biomedical Engineering Computing

[CS 1436](#) Programming Fundamentals

[CS 1325](#) Introduction to Programming

Project Experience (6 semester credit hours)

Select courses from the list below:

RIDE

[ECS 1192](#) RIDE Experience (may be repeatable twice)

[ECS 2192](#) Research, Inquiry, Design Experience (to be repeatable twice)

[ECS 3292](#) Research, Inquiry, Design Experience (to be repeatable twice)

[ECS 4392](#) Research, Inquiry, Design Experience (to be repeatable twice)

EPICS

[EPCS 2200](#) Engineering Projects in Community Service

[EPCS 3200](#) Engineering Projects in Community Service II

ENTP

[ENTP 3301](#) Innovation and Entrepreneurship

[ENTP 3320](#) Start-up Launch I

[ENTP 4350](#) Corporate Entrepreneurship

[ENTP 4360](#) Innovation and Creativity

And select any 16 semester credit hours from ECS at 2000 level or higher

Note: some major core courses may require the completion of additional prerequisites prior to enrollment.

Depth Field Requirement: 18 semester credit hours

Business Focus

Business Breadth (9 semester credit hours)

[BA 3300](#) Advanced Topics in Capitalism: Ideas and Challenges That Shape Our World

[BLAW 4301](#) International Law

[BLAW 4305](#) Introduction to Patent Law

[BPS 4307](#) Corporations, Politics and Society

[ENGY 3300](#) Introduction to Energy Technology

[ENGY 3301](#) Managing Carbon Assets: Oil, Gas, and Coal

[FIN 3300](#) Personal Finance

[MKT 3300](#) Principles of Marketing

[OBHR 3310](#) Organizational Behavior

[OBHR 3311](#) Principles of Management

[OPRE 3310](#) Supply Chain and Operations Management

[OPRE 3330](#) Project Management

Thematic Focus (9 semester credit hours)

Select 9 semester credit hours with a common theme (e.g., certificate, minor, foreign language)

Breadth Field Requirement: 19 semester credit hours

Select 9 semester credit hours from at least three different schools (outside of ECS)

Select 10 semester credit hours of elective courses

1. Semester credit hours fulfill the communication component of the Core Curriculum.
2. Three semester credit hours of Calculus are counted under Mathematics Core, and five semester credit hours of Calculus are counted as Component Area Option Core.
3. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted as Component Area Option Core.

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